Mining the Relationship Between Music Listening and Mental Health

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Abstract. Many individuals use music as a "medicine" in their everyday routine, with different functions for listening. Some use music to ease stress; others use music to help them focus; and some use it for relaxation (Thoma et al., 2013). This research will search for potential connections between the habits of music listening and the prevalence of anxiety, depression, and insomnia.
The study uses the Music and Mental Health survey dataset (which contains 736 participant responses with 33 different attributes per record). The data has the following attributes: time spent listening to music per day, preferred music genre, whether someone is an instrumentalist or composer, and subjective reports of participant mental health.
Using data mining techniques such as correlation analysis, predictive modelling, and clustering techniques, this research aims to look for observable patterns of relationships between music listening habits and mental health conditions. The results of this study will not produce a medical diagnosis but are intended to describe whether any patterns were found in the data (Chanda & Levitin, 2013).

1 Project Schedule and Timeline

Table 1. Project schedule and estimated effort.

Task	Time	Hours
Proposal and background reading	Jan 28 – Feb 12	15
Data cleaning and preprocessing	Feb 13 – Feb 19	8
Exploratory data analysis	Feb 20 – Feb 25	6
Feature engineering	Feb 26 – Mar 2	6
Classification model implementation	Mar 3 – Mar 12	10
Clustering analysis	Mar 13 – Mar 18	6
Model evaluation	Mar 19 – Mar 24	6
Result interpretation	Mar 25 – Mar 29	5
Draft report writing	Mar 30 – Apr 4	6
Visualization and refinement	Apr 5 – Apr 8	4
Presentation preparation	Apr 9 – Apr 11	4
Final report submission	Apr 12 – Apr 16	6

Gantt Chart

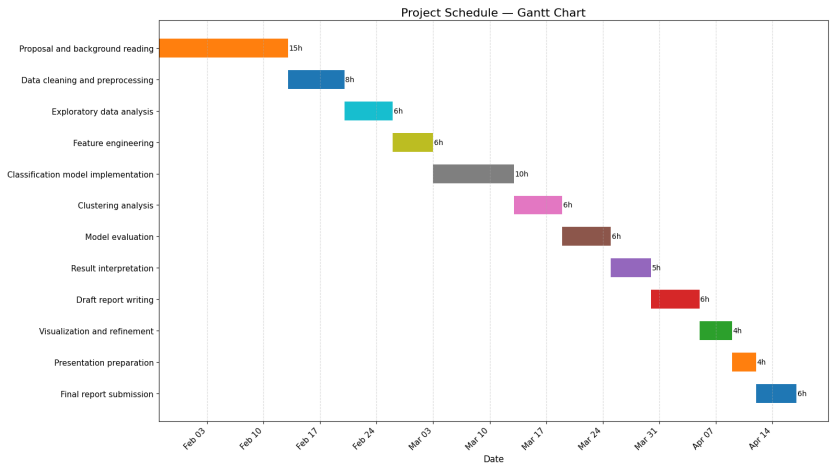

Fig. 1. Planned Gantt chart for the project timeline.

2 Dataset Description

The dataset I use is the Music & Mental Health Survey Dataset from Kaggle website. (https://www.kaggle.com/datasets/catherinerasgaitis/mxmh-survey-results)

The dataset includes:

- Age
- Primary streaming service used
- Hours of music listened per day
- While working (Do they listen while working?)
- Whether the person is an instrumentalist
- Whether the person is a composer or not
- Favorite music genre
- Frequency of listening to different genres (Rock, Pop, Classical, Hip-hop, EDM. . .)
- Exploratory listening behavior (Do they explore new music?)
- Foreign language music preference
- Whether music helps with focus
- Anxiety score
- Depression score
- Insomnia score
- Self-described music effects (positive, negative, no effect)

The dataset I downloaded is from Kaggle. It is in CSV format. Although 736 rows may not seem very large, I will focus on details and make sure everything is good. For classification models like Logistic Regression, KNN, SVM, and Random Forest, this size is suitable. Also, the dataset has enough features (33 columns) to allow meaningful analysis and feature prediction.

I also searched on many platforms (Jake provided on his website) for datasets related to music and mental health, and this one is the best and most suitable. I also did some research on GitHub, and overall, I saw only around 100 people who worked on similar topics before. 100 is not a small number, so I will explore and analyze this dataset from different perspectives and answer more questions using this dataset. I will try my best to complete this project with questions that no one has done before using this dataset.

3 Research Questions and Information to Mine

This project will explore the following questions:

1. Does listening to music relate to lower anxiety, depression, or insomnia levels?
2. Do instrumentalists or composers have better mental health than non-musicians?
3. What type of music is most related to mental health outcomes?
4. Is there a "best" number of hours per day to listen to music for better mental health?

The main goal is to find patterns in the data. I will explore:

- Correlations between listening hours and mental health scores
- Differences between musicians and non-musicians
- Genre-based differences in mental health
- Predictive models that classify mental health categories

The information mined from this dataset will be used to understand behavioral patterns. It will not be used for diagnosis or medical decisions. (Maybe I will add more thoughtful questions while doing the project; those four questions are fundamental questions.)

4 Why This Is Interesting and Useful

This project is interesting because many people believe that music affects mental health. Some people say music helps them relax. Others say sad music makes them feel worse. However, we want to see what the data shows.

From my own perspective, I came to Canada when I was 18 years old in 2022. I could not speak fluent English, and the new environment made me stressed. Because of my English ability, I could not make friends easily, so every day I was in a low mood until I found this "medicine" — music. I would listen to music on Spotify and YouTube when I was in a down mood, and this helped me a lot. Music made me feel much better, so I could learn English in a good mood and

adapt to the environment quickly. Because of this experience, and since I am doing my CS4403 project with Jake van der Laan, I see this as a great chance to work on this topic — finding the relationship between music and mental health.

This project connects daily behavior (music listening) with mental health indicators using data mining methods. From this project, I hope to learn:

- Whether music listening hours are positively or negatively related to mental health
- Whether musicians show different patterns compared to non-musicians
- Whether certain genres are linked to better or worse mental health scores
- Whether machine learning models can predict mental health categories from listening behavior

Even if the results show weak correlations, that is still useful. It helps us understand the limits of survey-based data and the difference between correlation and causation.

5 Methodology (my current idea so far, I may change it while doing the project)

The project will follow these steps:

Step 1: Data Cleaning

- Handle missing values
- Convert categorical variables (like genre) into numerical format
- Normalize numerical features
- Create new features if needed (low vs high depression category)

Step 2: Exploratory Data Analysis (EDA)

- Plot distributions of anxiety, depression, and insomnia
- Build correlation matrix
- Compare musician's vs non-musicians
- Analyze average scores by genre

Step 3: Supervised Learning

Models to test:

- Logistic Regression
- K-Nearest Neighbour (KNN)
- Support Vector Machine (SVM)
- Random Forest

Target variables:

- Depression category
- Anxiety category
- Insomnia category

Evaluation metrics:

- Accuracy
- Precision
- Recall
- F1 score

Step 4: Unsupervised Learning

- K-means clustering to find listener types
- Silhouette score to evaluate cluster quality
- Analyze cluster characteristics

Step 5: Feature Importance

- Identify which variables most influence mental health categories
- Compare model importance scores

6 Expected Results

I expect to find:

- Some correlation between listening hours and mental health scores
- Differences between musicians and non-musicians
- Genre-related patterns
- Possibly a moderate prediction accuracy for classification models

I do not expect extremely high prediction accuracy because mental health is influenced by many other factors not included in this dataset. And this project will focus on pattern discovery rather than strong medical conclusions.

7 Possible Challenges

- Self-reported data may contain bias
- Mental health is complex and cannot be fully explained by music
- Correlation does not mean causation
- Some genres may have fewer responses

These limitations will be clearly discussed in the final report.

References

1. Chanda, M. L., & Levitin, D. J.: The neurochemistry of music. *Trends in Cognitive Sciences* **17**(4), 179–193 (2013). https://doi.org/10.1016/j.tics.2013.02.007
2. Thoma, M. V., La Marca, R., Brönnimann, R., Finkel, L., Ehlert, U., & Nater, U. M.: The effect of music on the human stress response. *PLOS ONE* **8**(8), e70156 (2013). https://doi.org/10.1371/journal.pone.0070156